

# Formal Definition of the Limit

## $(\varepsilon - \delta$ Proofs)

### Contents

|          |                                                              |          |
|----------|--------------------------------------------------------------|----------|
| <b>1</b> | <b>Why Did We Need a Rigorous Definition?</b>                | <b>2</b> |
| <b>2</b> | <b>Intuition Behind Limits</b>                               | <b>2</b> |
| 2.1      | Visual Intuition . . . . .                                   | 3        |
| <b>3</b> | <b>Breaking Down the Definition</b>                          | <b>4</b> |
| 3.1      | $\varepsilon$ . . . . .                                      | 4        |
| 3.2      | $\delta$ . . . . .                                           | 4        |
| <b>4</b> | <b>How to write <math>\varepsilon - \delta</math> Proofs</b> | <b>4</b> |
| <b>5</b> | <b>Example 1</b>                                             | <b>5</b> |
| <b>6</b> | <b>Example 2</b>                                             | <b>5</b> |
| <b>7</b> | <b>Example 3: A Limit That Does NOT Exist</b>                | <b>6</b> |
| <b>8</b> | <b>Important</b>                                             | <b>7</b> |

# 1 Why Did We Need a Rigorous Definition?

Early calculus was developed by, Leibniz, and (though I disagree) Newton. They successfully developed derivatives, integrals, differential equations, and classical mechanics

However the problem was they didn't have rigorous definitions for everything they used. They used infinitesimals very loosely. This was not because they didn't try, but rather because they failed at giving a proper definition.

In this context, we mean quantities that are nonzero, but also infinitely small. Leibniz and Newton's concepts worked computationally, but, as any mathematical fact, we need to have precise definitions to make sure it does not break down, meaning we need a standard definition for everything.

This became a serious issue later on, and led to some difficult challenges. However, Cauchy and Weierstrass were able to rebuild calculus rigorously using limits. Their idea relied on replacing vague infinitesimals with inequalities. This is what leads us to the

$$\boxed{\varepsilon - \delta \text{ definitions}}$$

We can start off with the definition for non-infinitesimal limits.

We say:

$$\lim_{x \rightarrow c} f(x) = L$$

if for every:

$$\varepsilon > 0$$

there exists:

$$\delta > 0$$

such that whenever:

$$0 < |x - c| < \delta$$

it follows that:

$$|f(x) - L| < \varepsilon$$

Or more concisely:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

## 2 Intuition Behind Limits

Suppose we say:

$$\lim_{x \rightarrow 2} (3x + 1) = 7$$

What does this actually mean?

In highschool or any non real-analysis class, we are taught that when  $x$  gets close to 2, the output gets close to 7. As in, what value is the function approaching as  $x = 2$

Important:

- we do NOT care what happens exactly at  $x = 2$
- only what happens near  $x = 2$

## 2.1 Visual Intuition

We can think of  $|x - c|$  as the horizontal distance on the  $x$ -axis between  $x$  and  $c$ . Similarly, we may think of  $|f(x) - L|$  as the vertical distance on the  $y$ -axis between  $f(x)$  and  $L$ .

So, in simpler terms, the definition is saying:

if  $x$  is close enough to  $c$ , then  $f(x)$  must be close enough to  $L$ .

The goal is to make the vertical error:

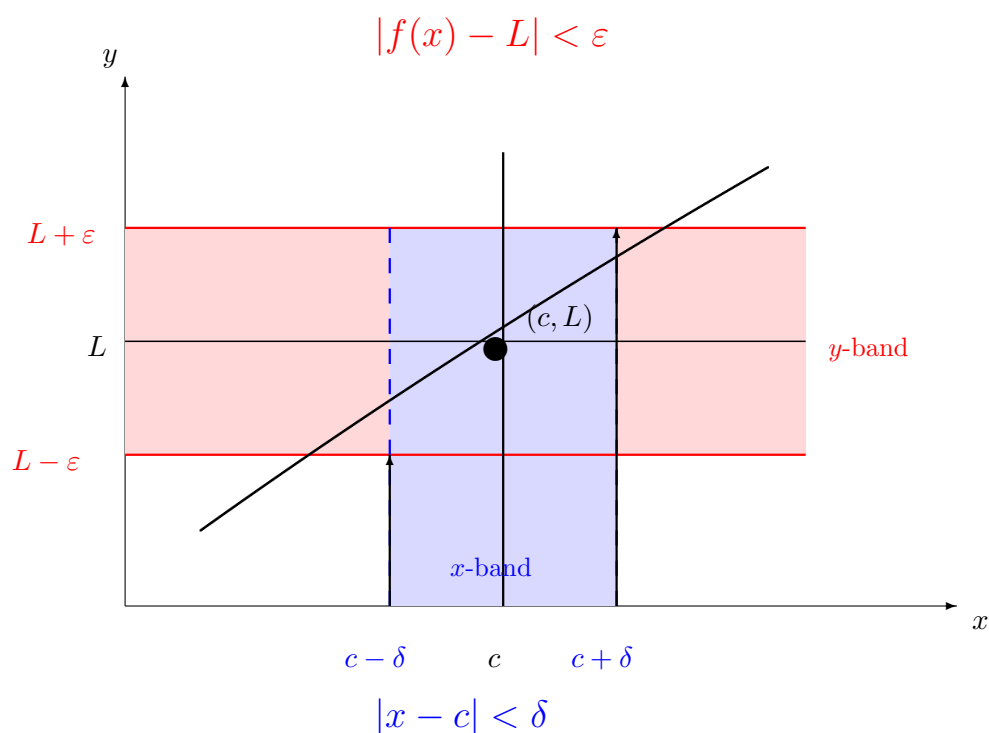
$$|f(x) - L|$$

as small as we want by forcing the horizontal error:

$$|x - c|$$

to be small enough. Once we make it small enough, we can keep making it infinitesimally smaller. There is no smallest upper bound on  $\epsilon$ , but there is however a largest upper bound. (Stangeby yaps for a million years about this, you'll get it).

In other words, the  $\delta$ -interval on the  $x$ -axis must map entirely into the  $\epsilon$ -interval on the  $y$ -axis. I've tried to draw a small visualization of the bounds for a random function.



You can interpret the picture as,  $c - \delta < x < c + \delta$  should force  $L - \varepsilon < f(x) < L + \varepsilon$

Which is just the first and second part of our definition, expanded out. In other words, no matter how small the  $\varepsilon$ -band becomes, we should be able to shrink the  $\delta$ -band enough to make the function stay inside it.

### 3 Breaking Down the Definition

#### 3.1 $\varepsilon$

$\varepsilon$  controls how close outputs must be to  $L$ .

Example:  $|f(x) - L| < 0.01$  This means  $f(x)$  must stay within 0.01 of  $L$ .

#### 3.2 $\delta$

$\delta$  controls how close our inputs must be to  $c$  such that our definition is valid. We are allowed to choose any  $\delta$ .

Important:

$\varepsilon$  is GIVEN to us

$\delta$  is OUR JOB to find

### 4 How to write $\varepsilon - \delta$ Proofs

1. Write:

$$|f(x) - L|$$

2. Write:

$$|x - c|$$

3. Algebraically manipulate both. Most of the time, our goal is to switch back to our  $|x - c|$  expression by manipulating  $|f(x) - L|$  to be some linear combination of the former.

4. Try to factor out:

$$|x - c|$$

5. It is often very useful to bound terms if you can't manipulate algebraically. A nice bound to try is always  $\delta = 1$

6. Choose  $\delta$  in terms of  $\varepsilon$ .

7. Finish the proof (reconstruct back to show the definition holds).

Usually:

$\delta = \min(\text{something}, \text{something})$

Those somethings, are your boundings you chose (if you chose any). You can think of it as one condition controls the algebra after bounding, and the other forces the inequality.

## 5 Example 1

Prove:

$$\lim_{x \rightarrow 2} (3x + 1) = 7$$

We need:

$$|(3x + 1) - 7| < \varepsilon$$

Simplify:

$$|3x - 6| = 3|x - 2|$$

With some rearranging, can now rewrite  $|f(x) - L|$  in the same exact way very as our  $|x - c|$  expression. Giving us:

$$|x - 2| < \frac{\varepsilon}{3}$$

Which means, we may choose:

$$\boxed{\delta = \frac{\varepsilon}{3}}$$

Done. (For the definition to be formally satisfied, you usually need to work back from  $\delta = \varepsilon/3$ )

## 6 Example 2

Prove:

$$\lim_{x \rightarrow 5} x^2 = 25$$

We need:

$$|x^2 - 25| < \varepsilon$$

Factor:

$$|x^2 - 25| = |x - 5||x + 5|$$

However, we still have a slight problem. We have no way to algebraically manipulate this into a linear combination of  $|x - 5|$ . Our  $x + 5$  expression still depends on  $x$ . We can now use bounding, which we talked about earlier. A very useful tool. We control this using a temporary restriction. Assume:

$$|x - 5| < 1$$

Then:

$$4 < x < 6$$

Therefore:

$$9 < x + 5 < 11$$

$$|x + 5| < 11$$

$$|x^2 - 25| < 11|x - 5|$$

Which we may now rewrite in terms of our  $|x - c|$  expression.

$$|x - 5| < \frac{\varepsilon}{11}$$

We can now choose:

$$\boxed{\delta = \min\left(1, \frac{\varepsilon}{11}\right)}$$

## 7 Example 3: A Limit That Does NOT Exist

Consider:

$$f(x) = \frac{1}{x}$$

We claim:

$$\lim_{x \rightarrow 0} \frac{1}{x}$$

does not exist.

Suppose, for contradiction, that:

$$\lim_{x \rightarrow 0} \frac{1}{x} = L$$

for some real number  $L$ .

Choose:

$$\varepsilon = 1$$

Then there exists  $\delta > 0$  such that:

$$0 < |x| < \delta \Rightarrow \left| \frac{1}{x} - L \right| < 1$$

Choose:

$$x_1 = \frac{\delta}{2} \quad x_2 = -\frac{\delta}{2}$$

Both satisfy:

$$0 < |x_1| < \delta \quad 0 < |x_2| < \delta$$

So:

$$\left| \frac{1}{x_1} - L \right| < 1 \quad \left| \frac{1}{x_2} - L \right| < 1$$

Substitute:

$$\left| \frac{2}{\delta} - L \right| < 1 \quad \left| -\frac{2}{\delta} - L \right| < 1$$

This means both values must be within 1 of  $L$ .

But their distance from each other is:

$$\left| \frac{2}{\delta} - \left( -\frac{2}{\delta} \right) \right| = \frac{4}{\delta}$$

Since  $\delta > 0$ ,

$$\frac{4}{\delta} > 1$$

So they cannot both be within 1 of the same number  $L$ .

Contradiction.

Therefore:

|                                                     |
|-----------------------------------------------------|
| $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist |
|-----------------------------------------------------|

## 8 Important

What's covered here is only the basics.  $\varepsilon - \delta$  proofs get much harder the more into real analysis you go. However this is everything you will need for this course.